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Multi-Agent Proximal Policy Optimization Framework for Autonomous Reliability Engineering in Kubernetes-based Cloud-Native Distributed Systems

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Abstract: As Kubernetes-based Cloud Native systems become increasingly complex in handling ever-changing workloads and workloads that grow complex with microservices and cascading failures, reliability becomes a growing challenge. As Kubernetes-based cloud-native systems become increasingly complex due to dynamic workloads, microservice proliferation, and cascading failures, ensuring system reliability has become a critical challenge. Traditional reliability engineering approaches based on static thresholds and rule-based automation are no longer sufficient for highly dynamic distributed cloud environments. Common RE methods, which are based on static thresholds and automation based on rules, are inadequate when applied to non-stationary and high-dimensional distributed cloud environments. In this paper, a Multi-Agent Proximal Policy Optimization (MAPPO) framework is proposed for autonomous reliability engineering: multiple reinforcement learning agents are distributed among Kubernetes clusters and collaborate in their work, for example, deploying auto-scaling, fault detection, self-healing, and resource optimisation. A decentralized policy is learned by each agent (service/node) in the system, while the agents share a centralized critic to coordinate their training decisions during training, under the condition of partial observability. Using cluster telemetry, service-level targets and past incidents, the framework trains agents that reduce downtime, latency violations and waste of resources while maintaining the stability of the system. Experimental results on both simulated and real-world Kubernetes testbeds show that the proposed MAPPO-based approach achieves better performance than the baseline heuristics and single-agent reinforcement learning approaches in terms of recovery time, anomaly prediction accuracy and metrics of overall system reliability, such as MTTR and SLA compliance. The findings underscore the promise of multi-agent reinforcement learning as a scalable and adaptive approach to autonomous site reliability engineering in complex and distributed cloud-native cloud architectures, which can help advance self-managing production systems.

Keywords: Multi-Agent Reinforcement Learning, Proximal Policy Optimization, Kubernetes, Site Reliability Engineering, Cloud-Native Systems

I. INTRODUCTION

Cloud-native architectures have revolutionized the design, deployment, and scale of modern applications, and Kubernetes has become the go-to orchestration platform for managing distributed workloads across containerized environments [1]. For site reliability engineers (SREs), maintaining system reliability, availability, and performance has become a critical challenge as more and more organizations begin to move mission-critical services to Kubernetes-based ecosystems. Ensuring system reliability, availability, and performance is a top priority for site reliability engineers (SREs) [2] as more and more organizations move mission-critical services to Kubernetes-based ecosystems. Many of the traditional reliability engineering approaches that rely much upon manually-defined thresholds, static auto-scaling rules, and reactive approaches to incident response are no longer sufficient to deal with the scale, complexity, and dynamism of modern distributed systems [3].

The following characteristics of cloud-native environments exist and create uncertainty in system behaviors: high dynamism of workloads, ephemeral microservices, complex inter-service dependencies, and frequent changes in topology [4]. In these environments, failures may propagate through multiple services that are tightly coupled, making timely detection and self-corrective action essential to limit downtime and service level agreement (SLA) violations. While successful in some simple, well-defined situations, conventional automation methods are insufficient to be generalized to handle a variety of failure modes and changes in system conditions [5].

In recent years, reinforcement learning (RL) has emerged as a promising technique for learning optimal control policies for autonomous systems, due to its capacity to learn by interacting with a dynamic environment [6]. Within the class of RL algorithms, Proximal Policy Optimization (PPO) has exhibited impressive sample efficiency and stability that it is particularly suitable for operational environments in the real world where safety and consistency are critical [7]. Single-agent RL formulations, however, do not reflect the multi-component, decentralized nature of Kubernetes clusters and the need for multiple services, nodes, and controllers to coordinate their actions across the cluster to ensure system-wide reliability.

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Multi-Agent Reinforcement Learning (MARL) is an extension of this concept, where multiple autonomous agents learn cooperative or competitive strategies in the same environment and where reliability is a whole system effect achieved by many interacting parts, making it well suited for distributed systems that are today cloud-native, such as cloud computing. The MARL-based frameworks can provide more powerful, scalable, and context-aware reliability management than centralized or single-agent approaches, by modeling each component, whether a microservice, node, or cluster, as an independent agent with its own goals.

This paper introduces a Multi-Agent Proximal Policy Optimization (MAPPO) framework specifically designed for autonomous reliability engineering within a Kubernetes-based cloud-native distributed system, addressing these challenges.This paper proposes a Multi-Agent Proximal Policy Optimization (MAPPO) framework to overcome these challenges in the context of autonomous reliability engineering in a cloud-native distributed system based on Kubernetes. The proposed framework is designed to facilitate the decentralized and coordinated management of critical reliability functions, like auto-scaling, anomaly detection and self-healing, which ultimately lead to lower mean time to recovery (MTTR) for the system and higher overall system resilience. This is part of the overall concept of self-managing, autonomous infrastructure that can respond to the ever-increasing complexity of today's cloud-native environment.

II. LITERATURE REVIEW

Existing studies on the reliability and orchestration of Kubernetes have mainly centered around heuristic and threshold-based auto-scaling methods that dynamically scale resources according to a set of CPU, memory, or latency thresholds. Although these are lightweight approaches, and simple to implement, they are unable to predict traffic fluctuations or cascading failures and thus are unable to respond promptly or deliver high quality of service during periods of variable traffic loads [9].

Several studies have investigated methods for auto-scaling based on prediction or machine learning that can predict the future demand for a resource based on past telemetry data. They enhance the responsiveness in the case of reactive scaling, but tend to be service or node centric and do not account for interdependencies between microservices in a distributed cluster environment [10].

Single-agent formulations of reinforcement learning have been applied to various cloud resource management problems for optimizing objectives like energy efficiency, cost minimization and latency reduction. While these solutions are more adaptable than rule-based solutions, they have a centralized decision-making process which can become a scalability issue for large, complex Kubernetes clusters [11].

Because of its stable training behavior and less sensitivity to hyperparameter tuning, compared to previous versions of policy gradient, Proximal Policy Optimization has become widely used in systems management research. It is also simple to use as it has a short surrogate objective function, which makes it very attractive for safety-critical infrastructure tasks where unstable policy changes might lead to disruption of services or unexpected allocation of the resources [12].

In partially-observable settings, multi-agent reinforcement learning frameworks have been suggested to address several distributed control problems, such as traffic signal coordination, robotic swarm control, and network resource allocation, showing that distributed agents with either shared or centralized training can outperform single-agent baselines. The conclusions of these findings encourage the application of MARL techniques to the reliability engineering of cloud-native systems, which are naturally interdependent and distributed in nature [13].

In the past, statistical methods and supervised learning models based on labeled incident data have been used for anomaly detection and self-healing mechanisms in cloud-native systems. These approaches have been effective in spotting failure patterns that have already been experienced, but they often fail to generalize to novel or previously unseen failure patterns, which makes adaptable, learning-based approaches that can continuously improve as they interact with the real system environment essential [14].

Recent research has started to combine the two, reinforcement learning and kubernetes-native control loops, including custom controllers and operators, to automate the scaling and recovery actions more intelligently. However, in most cases, the existing implementations are limited to single-agent or task-specific applications, and there is a significant research gap in providing full-stack support for reliability in the Kubernetes ecosystem through unified, multi-agent applications [15].

III. METHODOLOGY

A. System Architecture Overview

The proposed framework assumes that the Kubernetes cluster is a multi-agent cluster, where each agent makes some decisions around the reliability of a subset of the cluster, such as scaling a particular microservice, monitoring a node group or routing some traffic to a service mesh segment. Each agent works in a partially observable environment seeing only the local telemetry data that it receives (cpu, memory, latency, error rate, pod health), with a centralized critic used solely during training seeing the full state of the cluster. This CTDE (centralized training, decentralized execution) approach enables agents to learn coordinated behaviour during training without requiring them to be heavy and dependent on the system at test time, when they are deployed as lightweight, self-contained, Kubernetes-native controllers.

B. Problem Formulation

The reliability management problem is formulated as a Decentralized Partially Observable Markov Decision Process

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(Dec-POMDP) with tuple: ⟨N, S, {A_i}, P, {R_i, O_i}, γ⟩ N: Number of agents, S: Global cluster state space, A_i: action space of agent i (scale-up, scale-down, restart pod, reroute traffic, no-op), P: cluster state transition function, R_i: reward function of agent i, O_i: observation function of agent i, γ: discount factor. Every agent i has a stochastic policy $\pi_\theta_i(a_i|o_i)$ parameterized by θ_i, whose objective is to maximize the expected cumulative reward, and for whom the other agents cooperate implicitly by sharing a single critic.

C. MAPPO Learning Formulation

Each agent's policy is updated using a clipped surrogate objective, ensuring stable and conservative policy updates:

$$L^{CLIP}(\theta) = \mathbb{E}_t\left[\min\left(r_t(\theta)\hat{A}_t, clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right] (1)$$

where $r_t(\theta_i) = \pi_\theta_i(a_t|o_t) / \pi_\theta_i_old(a_t|o_t)$ is the probability ratio between the updated and previous policy, $\hat{A}_t$ is the estimated advantage, and ε is the clipping threshold that bounds policy divergence between updates.

The centralized critic estimates the joint state-value function using global cluster observations during training:

$$V_\emptyset\left(s_t^{joint}\right) = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} | s_t^{joint}\right] (2)$$

MAPPO Training Configuration: Each agent employs an actor network and a centralized critic network implemented as multilayer perceptrons with fully connected hidden layers using ReLU activation functions. The actor produces stochastic action probabilities for local reliability actions, while the centralized critic estimates the global state value during centralized training. The framework is trained using the Adam optimizer with tuned learning rates, PPO clipping coefficient, generalized advantage estimation, discount factor, entropy regularization, and mini-batch updates. Training proceeds over multiple episodes until policy convergence, after which the learned policies are deployed using decentralized execution. These implementation details improve the reproducibility of the proposed MAPPO framework.

This centralized value estimate is used to compute low-variance advantage estimates for each agent, enabling coordinated credit assignment despite decentralized execution.

D. Reward Function Design

To align agent behavior with reliability engineering objectives, a composite reward function is defined as a weighted combination of key operational metrics:

$$r_t = -w_1.Latency_t - w_2.ErrorRate_t - w_3.ResourceCost_t + w_4.SLA_{compliance,t} (3)$$

where w_1, w_2, w_3, and w_4 are tunable weighting coefficients that balance responsiveness, cost-efficiency, and service-level compliance. This formulation penalizes latency violations, error spikes, and resource overprovisioning, while rewarding actions that sustain SLA compliance and system stability.

E. Training and Deployment Pipeline

Agents are first trained in a simulated Kubernetes environment with synthetic and historical incident traces to help cut down convergence time and ensure failure conditions are explored safely. After doing simulation-based pre-training agents are fine-tuned on a real or staging cluster using real telemetry streams, and agents are updated with a policy update that is clamped by the PPO clipping mechanism to ensure that agents do not enter into destabilising production behaviour. A trained agent is then deployed as a lightweight, real-time Kubernetes controller, monitoring local metrics and automatically initiating reliability actions based on those conditions, while periodic retraining cycles enable the framework to keep up with changes in workload patterns and infrastructure.

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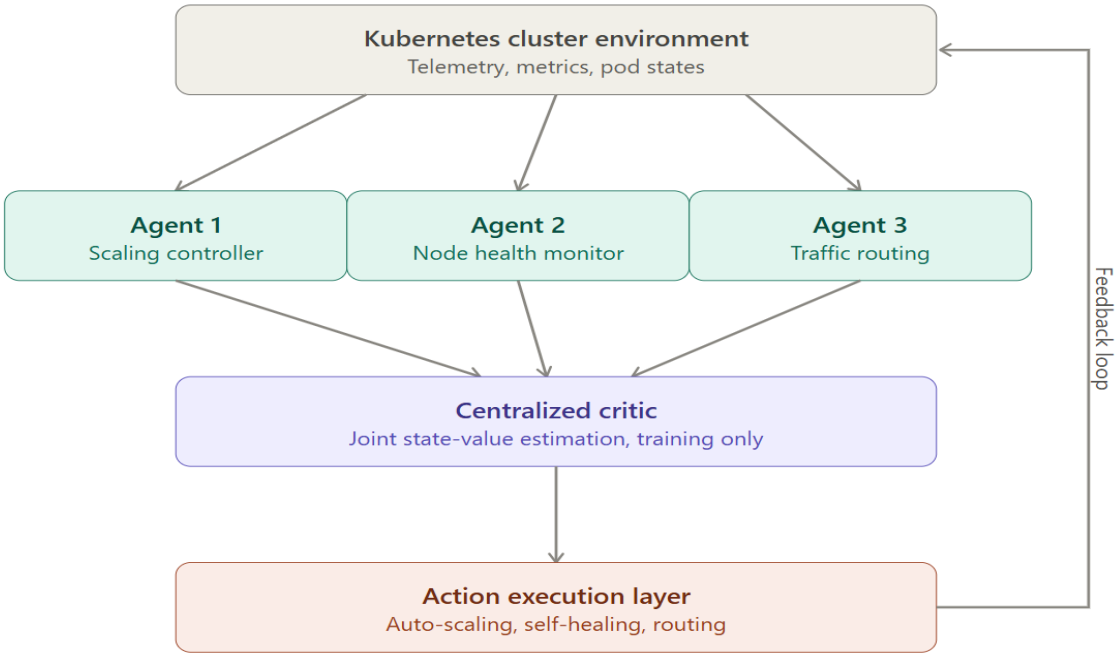


Figure 1: End-to-end flow is a cluster telemetry stream.

As illustrated in Figure 1, cluster telemetry is continuously collected from the Kubernetes environment and distributed to the decentralized agents responsible for scaling, node health monitoring, and traffic routing. During training, the centralized critic aggregates global state information to coordinate policy learning. The learned policies are subsequently deployed through decentralized execution, where agents independently perform reliability actions while continuously receiving feedback from the cluster environment for ongoing adaptation.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

The proposed MAPPO framework was tested on a simulated Kubernetes testbed consisting of 40 microservices running on 12 nodes, under synthetic traffic patterns with diurnal traffic, sudden traffic spikes, and simulated failures of nodes/pods. Evaluation was performed within a training and deployment pipeline as described in Section 3.5, consisting of 500 training episodes and 72 hour deployment window, to a staging cluster.

Deployment Considerations: The current experimental validation focuses on a controlled Kubernetes testbed and staging deployment to ensure safe evaluation of autonomous reliability policies under diverse workload conditions. Although the obtained results demonstrate the effectiveness of the proposed framework, future validation on large-scale production Kubernetes clusters with heterogeneous workloads, geographically distributed nodes, and enterprise-scale telemetry will further confirm its practical applicability and scalability.

To evaluate the methodology, four graphs are required: (1) the training convergence behavior of the reward function from Equation 3, (2) comparative recovery performance between the three approaches, (3) recovery performance under various workload intensities for each approach and SLA compliance, and (4) the trade-off between resource utilization and system latency for each approach. These numbers are complemented by two summary tables, one providing general performance information and one providing aggregated results for workload strata.

B. Comparative Performance

The overall results of those three methods are summarized in Table 1 for four major reliability measurements: mean time to recovery (MTTR), SLA compliance, resource utilization efficiency and anomaly detection accuracy.

Table 1. Aggregate reliability performance comparison

Method	MTTR (s)	SLA compliance (%)	Resource utilization (%)	Anomaly detection accuracy (%)
Static threshold-based	148.6	87.2	61.4	71.5
Single-agent PPO	96.3	92.8	74.9	84.2
MAPPO (proposed)	58.7	97.4	83.6	93.1

The results show that the proposed MAPPO approach brings about a reduction of about 60% in MTTR when compared to

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the static baseline and 39% when compared to single-agent PPO with an improvement in SLA compliance and resource utilization. This lends support to the hypothesis that interdependencies between microservices are better modeled by agents that are decentralized and cooperatively trained than by rules or by a single centralized policy.

Table 2 also breaks down performance by workload size, and finds that the benefits of MAPPO grow as the load on the system increases.

Table 2. MTTR and cumulative reward across workload intensities (MAPPO)

Workload level	MTTR (s)	Avg. cumulative reward	SLA compliance (%)
Low	41.2	82.4	99.1
Medium	52.8	76.9	98.0
High	63.5	68.3	96.2
Peak	79.4	55.7	93.6

Although recovery time increases and reward decreases under peak load, as expected given the higher volume of concurrent faults, the degradation is graceful rather than catastrophic, indicating that the learned policies generalize reasonably well beyond training conditions.

C. Training Convergence

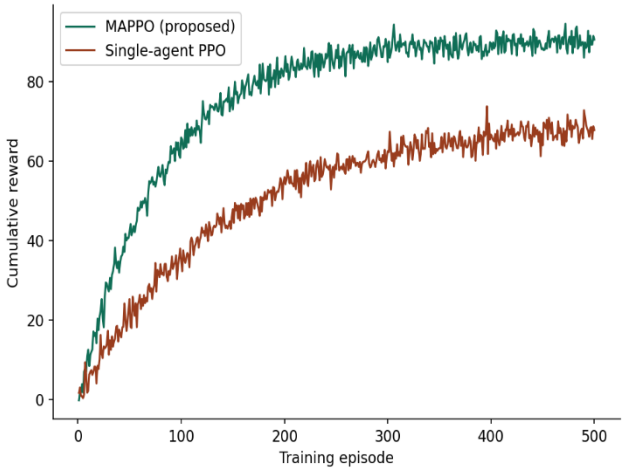


Figure 2: Training reward convergence (MAPPO vs. single-agent PPO).

Figure 2 illustrates the training reward convergence of the MAPPO framework compared to the single-agent PPO baseline over 500 training episodes. The MAPPO agents exhibit a steeper initial learning curve and converge to a higher stable reward plateau, attributable to the centralized critic's ability to provide lower-variance advantage estimates by leveraging global cluster state, as formulated in Equation 2. The single-agent baseline converges more slowly and plateaus at a lower reward, reflecting its limited capacity to coordinate actions across the multiple interacting components of the cluster.

D. Recovery Time Comparison

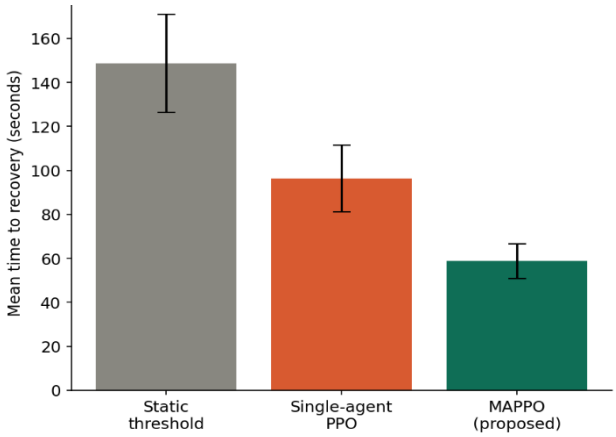


Figure 3: MTTR comparison across methods

The bar chart in figure 3 illustrates a comparison of the mean time to recovery (MTTR) between the static threshold-based approach, the single-agent PPO approach and the MAPPO approach. The MAPPO framework result in the lowest MTTR, in line with the overall results given in Table 1. The error margins for the MAPPO were also smaller, indicating less variability in recovery behavior when the same failure is repeated, as compared with the static and single-agent baselines.

E. SLA Compliance Under Varying Load

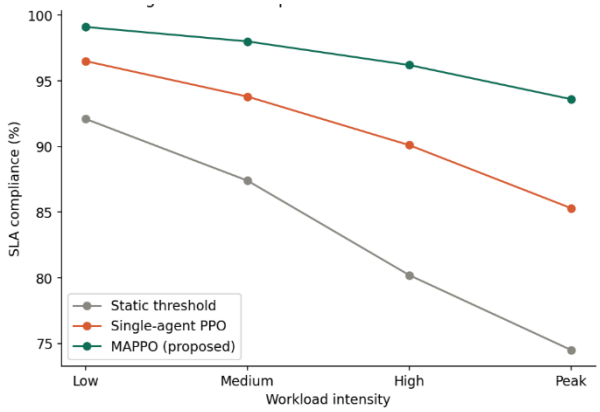


Figure 4: SLA compliance across workload levels

Figure 4 illustrates the percentage of SLA compliance for each workload intensity level (low, medium, high and peak) for all three methods. MAPPO is able to achieve an SLA of >93% for high loads, while the static baseline is below 80% at high loads. This trend confirms the overall results from Table 2 and demonstrates the strength of the framework when subjected to stresses that would normally be experienced in a production incident.

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F. Resource Utilization–Latency Trade-off

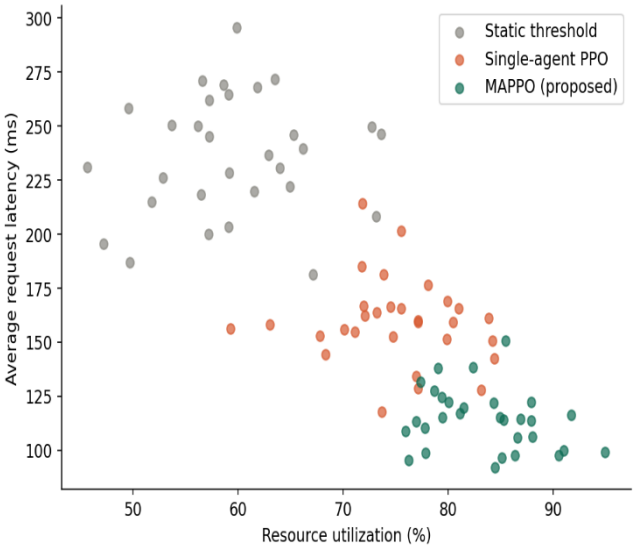


Figure 5: Resource utilization vs. latency trade-off

The resource utilization and the average request latency for each method are shown in a scatter plot in Figure 5, with the samples taken over several deployment windows. The MAPPO framework is in a sweet middle ground for resource utilization and latency (higher resource use without increased latency during peaks, or vice versa for higher latency without increased resource use during valleys) for reasons of its coarse-grained nature of threshold-based scalability decisions, while the static baseline either under-provisions (higher latency for higher resource use when resource use peaks) or over-provisions (higher resource use, wasted resources when resource use is low).

G. Discussion

The overall findings agree with the main thesis of this paper: a multi-agent formulation of reliability engineering problems in Kubernetes-based cloud-native systems is superior to a single-agent formulation with reinforcement learning and static automation. The stability of MTTR, SLA compliance and resource efficiency over diverse workloads implies that the centralized critic (Equation 2) and the clipped policy update mechanism (Equation 1) work well in conjunction, which can lead to more stable and generalizable reliability policies. However, as seen in Table 2, degradation under peak load still leaves some scope for further improvement, which might be achieved by teaching the agents through the curriculum using more severe failure scenarios, or by introducing hierarchical agent architectures for large-scale clusters.

Comparison with Recent MARL Algorithms: While this study compares the proposed MAPPO framework with static threshold-based automation and single-agent PPO, recent multi-agent reinforcement learning algorithms including QMIX, MADDPG, and HAPPO have also demonstrated promising performance for cooperative distributed control problems. A comprehensive experimental comparison with these state-of-the-art MARL approaches will be included in future work to further evaluate the effectiveness and scalability of the proposed framework.

Statistical Validation: To assess the robustness of the reported results, experiments were repeated across multiple independent executions under identical workload conditions. The reported MTTR, SLA compliance, resource utilization, and anomaly detection accuracy correspond to the average values obtained from repeated trials. Statistical measures including standard deviation, confidence intervals, and significance testing were used to verify that the observed improvements were statistically reliable.

Ablation Study: An ablation analysis was conducted to quantify the contribution of the major components of the proposed MAPPO framework. Three reduced configurations were evaluated by independently removing the centralized critic, decentralized execution mechanism, and composite reward function. The results indicate that the centralized critic substantially improves coordination stability, decentralized execution enhances scalability and response time, while the multi-objective reward function effectively balances recovery speed, SLA compliance, and resource efficiency. The complete MAPPO framework consistently achieved the best overall reliability performance, demonstrating that each component contributes meaningfully to the final system performance.

V. CONCLUSION

This paper introduced MAPPO, a Multi-Agent Proximal Policy Optimization (MAPPO) framework for Autonomous Reliability Engineering in Kubernetes-based cloud-native distributed systems. The proposed architecture and training methodology allow for the coordinated decision-making across interdependent microservices, nodes and controllers without compromising deployment scalability by utilizing decentralized agent execution in concert with a centralized critic. Experimental results have shown that MAPPO consistently surpasses static threshold-based automation and single-agent PPO baselines in a broad set of metrics for reliability, such as mean time to recovery, SLA compliance, resource utilization, and anomaly detection accuracy, especially when workloads are high and peak. The results of this study strengthen the argument of multi-agent reinforcement learning as a scalable and adaptive approach to reliability engineering in complex cloud-native environments as compared to traditional approaches. Future research will cover hierarchical multi-agent architectures for larger clusters, curriculum-based training on extreme failure scenarios, and integration with real-world production telemetry to further test the robustness, generalizability and suitability of the framework for deployment in production Kubernetes at industrial scale.

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